Vienna, November 6, 2026

Dr. Bill Munro
Associate Editor, Physical Review A

Re: XU11228A, "Answer partitions and oracle access determine quantum query complexity" by Karl Svozil
(revised title: "Answer Partitions, Oracle Response Spectra, and Exact Quantum Query Advantage")

Dear Dr. Munro,

I thank both referees for careful reports. Both ask for a clearer separation between known and new results, and the second referee raises the numerical status of part of the four-bit census. I have revised the manuscript accordingly. The main changes are:

1. **The four-bit census is now exact.** For the 43 orbit representatives (6040 partitions) that were previously supported only by numerical SDP infeasibility, the revised manuscript gives rational dual certificates of the two-query Barnum–Saks–Szegedy program. Each certificate bounds the optimal two-query worst-case success by a value between 0.9001 and 0.9862, which proves $Q_E = 3$. The certificates are lists of integers verified in exact arithmetic by a standalone verifier included in the supplement; the validity argument is a three-line weak-duality chain, Eq. (17). All 6435 balanced four-bit partitions are therefore classified exactly (Table III).

2. **Prior work versus new results.** A new Table I lists, topic by topic, what was known and what is added. The introduction now names the main new result (the response-spectrum criterion, Theorem 1, which contains Deutsch's problem as Proposition 4) and the second (the exact four-bit classification), and states that Propositions 1 and 2 are standard.

3. **Three-bit values credited.** The abstract, the caption of Table II, and Sec. IV now state that the exact three-bit complexities and the majority construction are those of Montanaro, Jozsa, and Mitchison.

4. **Bounded error.** New Proposition 5 treats odd cyclic responses with bounded error, includes the second referee's construction, and proves that the improved one-query success $(3 + \cos\frac{\pi}{d})/4$ is optimal.

5. **Access conventions.** A new paragraph before Proposition 4 (Theorem 1 of the submitted version) states the access assumptions and the phase convention.

6. **Generalization and structure.** The response-spectrum theorem of the submitted version is extended to a complete classification of the one-query-exact total Boolean functions for every response unitary on any number of bits. This classification is now Theorem 1; the submitted Theorem 1, Deutsch's problem, is its special case, Proposition 4. For the XOR oracle the classification reduces to the characterization of Chen, Ye, and Li [Phys. Rev. A **101**, 022325 (2020)]. The cost of identification with an inaccessible witness is now determined exactly (Proposition 6). The manuscript is now organized as a regular article with sections and numbered statements, and the title names the main result. To make the paper easier to follow, the strategy common to all results is now stated in four steps in the Introduction and recalled in the Discussion (Sec. VI).

Point-by-point replies follow. Section, equation, table, and statement numbers refer to the revised manuscript. Because the result on Deutsch's problem is now a special case of a more general theorem,

two statements have been renumbered: Theorem 1 of the submitted version, the response-spectrum theorem to which the reports refer, is now Proposition 4, and Theorem 1 is now its generalization to every number of bits. Quotations from the reports keep the original numbering. A list of changes is given at the end, and a marked-up PDF is attached. References to "Ref. [n] of the submitted version" use the numbering of the submitted manuscript, as in the reports.

Reply to the First Referee

> *I am not very clear about the contribution relative to prior work. The manuscript should state more explicitly what is already known and what is established for the first time here, particularly in relation to Refs. [13] and [14]. [...] A concise comparison paragraph or a small table distinguishing prior results from the present additions would be sufficient.*

I agree, and I have added both. The introduction now contains a paragraph naming the new results, and Table I separates prior results from the present additions. With respect to the two references named:

- Montanaro, Jozsa, and Mitchison (Ref. [13] of the submitted version) give explicit optimal exact algorithms for every function on up to three bits, and numerical SDP values for all functions on four bits, which they note do not by themselves certify exactness. The revised manuscript states that the three-bit values in Table II are theirs and adds no new three-bit values. For four bits it adds exact rational dual certificates for every balanced partition whose two-query infeasibility was previously only numerical (Sec. IV, Table III).
- Ye, Yang, and Huang (Ref. [14] of the submitted version) study exact two-query algorithms using the four-bit numerical classification. The present certificates make the balanced part of that classification rigorous.

The results that are new, to my knowledge, are the response-spectrum criterion for an arbitrary response unitary (Theorem 1, with Deutsch's problem as Proposition 4), its consequences for qutrit oracles with the same classical values (Corollary 1), the bounded-error statement for odd cyclic responses (Proposition 5), the exact four-bit certificates, and the exact costs of the witness channel (Proposition 6). Propositions 1 and 2 are standard discrimination conditions and convex duality, stated in the form used later.

Reply to the Second Referee

> *(1) [...] Ref. [13] [...] also provides explicit optimal exact algorithms for all Boolean functions on up to three bits. Its Sec. 6.2 includes the same majority construction described here. Thus, the exact three-bit query complexities are already known. The author should acknowledge this directly when presenting Table I and in the abstract.*

The referee is right. The abstract now says that the paper reorganizes the known exact complexities of the 35 balanced three-bit partitions. The caption of the three-bit table (now Table II) states that its values are given in Ref. [13], and the text in Sec. IV says that the contribution for three bits is

only their organization as balanced partitions and that the majority algorithm is the one in Sec. 6.2 of Ref. [13].

> *(2) More generally, the manuscript should identify its main original scientific result earlier. The dependence of query complexity on the oracle interface is already discussed in Ref. [1]. Proposition 1 is also acknowledged to follow from standard discrimination conditions. The response-spectrum theorem appears to be the most concrete contribution. Its significance would be easier to assess if the introduction clearly distinguished this result from the background examples.*

I agree. The introduction now states that the dependence on the oracle model is known in general, citing Ref. [1] (Wójcik and Chhajlany) together with Kashefi *et al.*, Calude, and Regev and Schiff. A separate paragraph, beginning "The main new result is a spectral criterion", names the response-spectrum criterion (Theorem 1) as the main new result, identifies the exact four-bit classification as the second, and describes Propositions 1 and 2 as framework. Table I makes the same separation item by item.

The response-spectrum result has also been strengthened. Theorem 1 now shows that for point access through any response unitary V on any number of bits, one query resolves a dictator exactly if and only if 0 lies in the convex hull of the spectrum of V, and a two-bit parity if and only if -1 is an eigenvalue of V; no other total Boolean function is one-query exact. For the XOR oracle this is the theorem of Chen, Ye, and Li; for an odd cyclic response the one-query class reduces to the dictators; and for $V = -I$, the sign oracle, it contains the two-bit parities but no dictator. Every response with the classical readout $V|0\rangle \propto |1\rangle$ meets the dictator condition, so for such oracles the spectrum decides only the two-bit parities. The proof for Deutsch's problem (Proposition 4) now uses an explicit dual witness, $Z = 2I_2 \otimes (I+V)(I+V)^\dagger$ [Eq. (11)], which is positive definite exactly when $-1 \notin \operatorname{spec} V$.

> *(3) The even-odd distinction in Corollary 1 concerns exact computation. A short discussion of bounded-error computation would help clarify its significance. [. . .] Hence one query still gives a bounded-error quantum advantage [. . .]. Please discuss this distinction.*

I thank the referee for this point; it is now part of the paper. Section III reproduces the referee's construction, credited to an anonymous referee, (success 1 on constant and 3/4 on balanced instances, against 1/2 for one classical query) and then proves a sharper statement, Proposition 5. For odd d, a probe spread equally over the two eigenvectors of X_d whose eigenvalues are adjacent to -1 solves Deutsch's problem with one query and zero error. The answer is conclusive with probability $(1+\cos\frac{\pi}{d})/2$ on every instance, and guessing on the inconclusive outcome gives worst-case success $(3+\cos\frac{\pi}{d})/4$. For $d = 3$ these values are 3/4 and 7/8. The value is optimal: for any probe, the cell-averaged output states differ only in an off-diagonal block bounded by $\|I - V\|^2$, which limits every one-query protocol to success $\frac{1}{2} + \frac{1}{8}\|I - V\|^2$, equal to $(3+\cos\frac{\pi}{d})/4$ for odd d. The same bound reaches 1 only when -1 is an eigenvalue of V, which gives a second proof of Proposition 4. One classical query never yields a conclusive zero-error answer. The text now says explicitly that the odd cyclic response removes the exact advantage but not the bounded-error one, and the abstract mentions the 7/8 value.

> *(4) The access assumptions underlying Theorem 1 should be stated explicitly. [. . .] the sufficiency argument assumes that an appropriate eigenstate of the chosen V can be prepared at no query cost. Presumably, V is known, and arbitrary, instance-independent*

operations are allowed. There is also a phase convention worth explaining. Replacing V by $e^{i\theta}V$ can change the condition in the theorem. In Eq. (9), this phase becomes relative between address sectors and can therefore be observable. [...]

The referee's reading is correct, and the assumptions are now stated in a paragraph "Access conventions" directly after the definition of the controlled-response family [now Eq. (10)]. The family is specified and V is known; any instance-independent operation, including the preparation of eigenvectors of V, is free; the oracle may not be controlled or inverted unless that access is specified separately. The paragraph explains that $e^{i\theta}V$ multiplies the oracle by the instance-dependent phase $e^{i\theta b_x}$, which is relative between address sectors and observable, and that this distinguishes the specified oracle from access to an unspecified unitary or channel known only up to a global phase. Corollary 1 now uses the phase convention as an example: X_3 and $e^{i\pi/3}X_3$ act identically on every computational-basis input up to a phase, yet the second has eigenvalue -1 and gives $Q_E = 1$. The corollary also compares two reversible qutrit oracles that agree on the blank target $|0\rangle$, the transposition $|0\rangle \leftrightarrow |1\rangle$ (spectrum $\{1, 1, -1\}$, $Q_E = 1$) and the cyclic shift ($Q_E = 2$).

(5) The numerical status should be reflected consistently in the supplementary data. For the 43 unresolved four-bit orbits, the CSV column "Q_lower_bound" and the JSON field "lower_bound" contain the value 3. However, the rigorous bounds supplied for these cases are $2 \le Q_E \le 3$. [...] Please separate the proved lower bound from the numerical candidate. Exact certificates would strengthen the census, but repeating SCS at a tighter tolerance does not establish these missing lower bounds, as the manuscript correctly notes.

The referee was right that the submitted supplement mixed proved and numerical bounds. Following the suggestion, the missing lower bounds are now proved by exact certificates rather than by further numerics. I used the bounded-error form of the two-query SDP, whose dual is well posed, rather than an exact-feasibility test. To a numerical dual solution I added 3ε, 2ε, and ε times the identity to Y_1, Y_2, and Y_3 ($\varepsilon = 2^{-20}$), which raises every constraint matrix by εI, and rounded all entries to dyadic rationals with denominator 2^{40}; all twelve matrix inequalities and the conditions on λ were then verified in exact integer arithmetic. All 43 certificates are valid, so $Q_E = 3$ for these orbits is now a theorem (Sec. IV, Table III). Their numerical optima, 0.9000–0.9861, agree with the values reported by Montanaro, Jozsa, and Mitchison, and the 15 orbits settled by trees give dual value 1, as they must.

The supplement is now submitted as APS Supplemental Material. In its principal files (`data/results-m4-orbits.csv`, `witnesses-m4/`, and the cross-check `data/m4-mjm-crosscheck.csv`) the 43 rows formerly labelled `numerical_candidate` now carry `exact_dual_certificate`, and every row states the source of its lower bound in a new field (`Q_lower_bound_source`: trivial, the exact one-query obstruction, or a rational dual certificate), together with the certified two-query success bound and the location of the certificate. A new folder `dual-certificates/` contains the 43 certificates, keyed by the partition numbers of the CSV file, and a verifier that uses only the Python standard library. The raw SCS rerun files are kept unchanged as the numerical record; a note in that folder states that their lower bounds were numerical when written and are superseded by the certificates.

(6) A few presentation details should be corrected. The supplementary PDF and README use a different manuscript title. The notation D should be explicitly related to the previously introduced classical quantity R_0. The term "kinematic partition" is introduced but does not seem necessary for the subsequent analysis.

All three are corrected. The supplementary PDF, the README, and the Lean documentation now carry the revised manuscript title and mention the title under which it was submitted. Section II now states that $D = R_0$ in the worst-case convention used in the paper, because fixing the random string of a zero-error protocol with a worst-case query bound leaves a deterministic protocol with the same bound, and notes that zero-error cost measured in expectation can be smaller. The term "kinematic partition" has been removed.

List of changes

- **Title** changed to "Answer Partitions, Oracle Response Spectra, and Exact Quantum Query Advantage".
- **Abstract** rewritten: main result first; three-bit values described as known; four-bit classification exact; bounded-error value 7/8 added; the one-query classification is stated for total Boolean functions, and dictators are defined in words.
- **Format**: `pra` class, sections, theorem environments with proofs.
- **Language**: the text has been revised throughout to state things more directly and readably; no result, equation, or statement was changed by this revision. Every nontrivial symbol is now defined where it first appears, and the term "oracle" is explained once, with reference to Turing's oracle machines, and then used throughout.
- **Sec. I (Introduction)**: first two paragraphs rewritten to explain the XOR oracle and why two qutrit oracles with the same classical readout can differ in quantum cost; prior work on oracle-model dependence cited (Wójcik–Chhajlany, Kashefi *et al.*, Calude, Regev–Schiff); relation to earlier state-partition work stated; new paragraph stating the four-step strategy used throughout the paper, to make the argument easier to follow; paragraph naming the new results; new Table I (prior work versus this work); closing paragraph on quantum parallelism shortened, since the new summary covers its first part.
- **Sec. II**: term "kinematic partition" removed; D defined and related to R_0; Proposition 1 with proof; new Proposition 2 (dual certificate) and its relation to adversary progress measures; remark on commuting oracle families.
- **Sec. III**: Deutsch citation corrected to include the exact one-query version (Cleve *et al.*); new Proposition 3 (sign oracle cannot resolve partitions that separate a truth table from its complement); access-conventions paragraph; Theorem 1 of the submitted version (Deutsch's problem) renumbered Proposition 4 and proved with an explicit dual witness, with the sum of the dual terms written out; Corollary 1 rewritten and retitled (transposition versus cyclic shift, shift with an added phase, general X_d with $d \geq 2$); new Theorem 1 (complete one-query classification of total Boolean functions for every response unitary), with a remark that promise problems such as Deutsch–Jozsa lie outside it, a remark that every response with the classical readout $V|0\rangle \propto |1\rangle$ meets the dictator condition, and with the equality case used in its proof stated explicitly; new Proposition 5 (optimal bounded-error solution for odd cyclic responses, $d \geq 3$), with the referee's construction and a remark that it is also optimal among unambiguous protocols; parity paragraph with explicit dual witness, degree bound for every response, and an open question.
- **Sec. IV**: three-bit values credited to Montanaro–Jozsa–Mitchison (caption of Table II and text); truth-table bits in Table II written b_i, as in the text; one-query lower bounds from Theorem 1;

two-query lower bounds from rational dual certificates [Eqs. (16)–(17)]; new Table III for four bits; all counts now exact (4847 partitions with an exact advantage, 1588 without); real-degree remark; new paragraph describing the Lean 4 formalization in the supplement and its trust assumptions.

- **Sec. V**: witness channel written as dephasing followed by the oracle; register of the reduced state stated; new Proposition 6 (exact identification cost $\min\{k+1, n\}$ with a witness of k address bits); new closing remark on partially distinguishable witnesses with overlaps $\kappa_i < 1$: Proposition 6 holds unchanged, so any leakage removes the exact one-query advantage, while one query still identifies the address with bounded error when $\prod_{i \in S}(1 + \kappa_i) > 2^{k-1}$.

- **Sec. VI (Discussion)**: new opening paragraph on whether a classical algorithm can ever need fewer calls. In the standard models and in every access model of the paper it cannot, though the advantage can vanish (qutrit shift, parity, Grover optimality). Then a short paragraph recalling the four steps of the Introduction, with a remark that with more than two branches phases other than -1 can cancel (as in the quantum Fourier transform, and for the shift X_3 in the dictator condition of Theorem 1), and a remark on several calls and bounded error. A later paragraph says which results vary which step, and separates the sign oracle, whose values are invisible to basis queries, from the examples that keep the classical values fixed. New remark that the Deutsch and Bernstein–Vazirani separations use only Clifford operations, so stabilizer simulability and query advantage are distinct (with Gottesman, Aaronson–Gottesman, and Veitch *et al.*); Shor and Grover cited; asymptotic remark sharpened; a second open question (which response is optimal for several queries); and a closing paragraph that summarizes what the paper adds and states that all effects found are constant factors.

- **References**: added Chen–Ye–Li, Ye–Li, Kashefi *et al.*, Calude, Regev–Schiff, Ambainis (adversary), Cleve *et al.*, Deutsch–Jozsa, Naseri *et al.*, Gottesman, Aaronson–Gottesman, Veitch *et al.*, Shor, Grover, Bennett *et al.* (optimality of Grover search), Turing (oracle machines), and three earlier papers on state partitions; Aaronson's open-problems survey removed.

- **Use of AI tools** disclosed according to the APS policy: tools and versions, the tasks they assisted with, and how the results were verified, in the Acknowledgments and, for the computations, in the Methods paragraph of Sec. IV.

- **Data availability**: the supplement is now provided as APS Supplemental Material, with the standard citation in the manuscript.

- **Supplemental Material**: new folder with the rational dual certificates and their exact verifier; lower-bound fields labeled by source, and numerical-candidate labels replaced, in the principal files; new section on the certificates in the supplementary PDF; title corrected; the term "kinematic" removed from the Lean sources.

Sincerely,

Karl Svozil